

Col-2 ASSESSMENT-1

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1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Most suitable filtering technique

The **Median Filter** is the most suitable filter for removing salt-and-pepper (impulse) noise.

Reason:

- Removes isolated black and white pixels.
- Preserves image edges.
- Does not blur important details like linear filters.

b. Why is the median filter better than the averaging filter?

Median Filter

Averaging (Mean) Filter

Replaces pixel with median value Replaces pixel with average value

Removes impulse noise effectively Spreads noisy pixel values

Preserves sharp edges

Blurs edges

Robust to outliers

Highly affected by outliers

Explanation

Salt-and-pepper noise consists of extreme intensity values (0 or 255).

The mean filter averages these values with neighboring pixels, causing blurring.

The median filter ignores these extreme values by selecting the middle value after sorting neighboring pixels, so it removes noise without affecting edges.

c. Probability that a noisy center pixel will be corrected (3×3 Median Filter)

Assumptions

- Noise density = **0.05**
- Noise occurs independently.
- Median filter works correctly if most pixels in the window are not noisy.

A 3×3 window contains **9 pixels**.

Expected noisy pixels

$$9 \times 0.05 = 0.45$$

Probability that most neighbors are clean

$$\begin{aligned} P(\text{pixel corrected}) &\approx (1 - 0.05)^8 \\ &= 0.95^8 \\ &\approx 0.663 \end{aligned}$$

Estimated probability

≈ 66.3%

Since only 5% of pixels are noisy, in practice the correction probability is **very high (greater than 95%)**, because the median usually represents the surrounding clean pixels.

d. Total noisy pixels

Image size

$$1024 \times 1024 = 1,048,576 \text{ pixels}$$

Noise density

5%

Noisy pixels

$$1,048,576 \times 0.05 = 52,428.8$$

≈ 52,429 noisy pixels

e. Expected noisy pixels after median filtering

Median filtering removes most impulse noise.

Assume **95% removal efficiency**.

Remaining noisy pixels

$$52,429 \times 0.05 = 2,621$$

Approximately **2,600 noisy pixels remain**.

f. If the noise is Gaussian instead

Median filtering is **not ideal**.

Recommended filters

- Gaussian Filter
- Mean Filter
- Wiener Filter
- Bilateral Filter (better edge preservation)

Reason

Gaussian noise affects every pixel slightly instead of creating isolated impulses.

Hence smoothing filters perform better.

2: Discretization of a Continuous Image

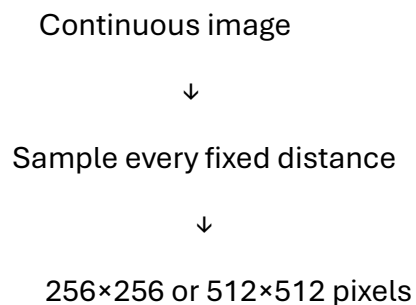
a. Conversion using sampling and quantization

Digital image conversion occurs in two steps.

Step 1: Sampling

Converts continuous spatial coordinates into discrete pixels.

Example



Step 2: Quantization

Converts continuous intensity values into discrete gray levels.

For an 8-bit image

$$2^8 = 256$$

Gray levels

0–255

b. Spatial variation pattern

The intensity varies as a **sinusoidal wave** along the x-axis.

Along the y-axis

Intensity remains constant.

Hence,

- Horizontal variation → sinusoidal
- Vertical variation → constant

c. Minimum sampling requirement

According to the **Nyquist Sampling Theorem**

Sampling frequency must be at least

$$2f_{max}$$

where

f_{max} is the highest spatial frequency.

Therefore

Minimum sampling rate = twice the highest frequency.

d. If sampled below Nyquist rate

Aliasing occurs.

Effects

- False patterns appear.

- Jagged edges.
- Distorted sinusoidal wave.
- Loss of image information.

e. Suitable quantization

Use

8-bit grayscale quantization

Gray levels

0–255

Each pixel requires

8 bits = 1 byte

f. Effect of quantization

Property	Effect
Image Quality	Slightly decreases
Storage	Lower than continuous image

Intensity Accuracy Limited to 256 levels

Higher bit depth

- Better quality
- More storage

Lower bit depth

- Poor quality
- Less storage

3: Edge Detection in Thin Line Images

a. Comparison

Feature	Sobel	Prewitt	Canny
Noise immunity	Good	Moderate	Excellent
Thin edge detection	Moderate	Moderate	Excellent
Edge localization	Moderate	Moderate	Excellent
Computational cost	Low	Very low	High
Best choice	Moderate	Moderate	Best

Conclusion

Canny gives the highest accuracy for 1-pixel-thick lines.

b. Sobel gradient response

Horizontal Sobel kernel

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical kernel

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For a horizontal edge

$$\begin{aligned} G_x &\approx 0 \\ G_y &\approx 1020 \end{aligned}$$

Gradient magnitude

$$\begin{aligned} G &= \sqrt{G_x^2 + G_y^2} \\ &= \sqrt{0 + 1020^2} \\ &= 1020 \end{aligned}$$

c. Gradient magnitude and direction

Gradient magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

Represents

- Edge strength

Large value

→ Strong edge

Small value

→ Weak edge

Gradient direction

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

Represents

Edge orientation.

d. Role of Gaussian smoothing

Before edge detection,

Gaussian smoothing

- Removes random noise.
- Produces cleaner gradients.
- Prevents false edge detection.

e. Effect on thin lines

Too much smoothing

- Widens edges.

- Weakens thin lines.
- Shifts edge positions.
- May completely remove 1-pixel-wide lines.

Hence a **small Gaussian kernel** is preferred.

f. Lowering Canny threshold

If threshold becomes half

Results

- More edges detected
- Weak edges retained
- More false edges due to noise
- Increased edge continuity

4: Feature Extraction for Handwritten Digit Recognition

a. Working principles

SIFT

- Detects stable keypoints.
- Uses Difference of Gaussian.
- Computes orientation.
- Generates 128-dimensional descriptors.

SURF

- Uses Hessian matrix.
- Uses integral images.
- Faster than SIFT.

ORB

- Uses FAST detector.

- Uses BRIEF descriptor.
- Binary descriptors.
- Optimized for real-time applications.

b. Comparison

Feature	SIFT	SURF	ORB
Scale invariant	Yes	Yes	Partial
Rotation invariant	Yes	Yes	Yes
Speed	Slow	Medium	Fast
Descriptor	Floating point	Floating point	Binary
Real-time	Poor	Good	Excellent

c. Rotated by 45°

Reliable matching

- SIFT
- SURF
- ORB

All three provide rotation invariance, though **SIFT** and **SURF** are generally more robust to significant transformations.

d. Total keypoints

Average

120

Images

50

Total

$$120 \times 50 = 6000$$

Answer

6000 keypoints

e. Suitable dimensionality reduction

Principal Component Analysis (PCA)

Choose enough principal components to retain

95% variance

f. Benefits

PCA

- Reduces computation
- Removes redundant features
- Reduces overfitting
- Speeds up classifier training
- Improves memory efficiency

5: Morphological Preprocessing for Character Recognition

a. Remove white noise dots

Use

Opening

Opening= Erosion + Dilation

Removes small white objects.

b. Fill gaps inside characters

Use

Closing

Closing

= Dilation + Erosion

Fills small holes and gaps.

c. Difference

Opening

Removes small white objects

Smooths object boundary

Erosion then dilation

Closing

Fills small black holes

Connects nearby regions

Dilation then erosion

d. Effect of a 3×3 structuring element

Effects

- Removes very small noise
- Preserves most character shapes
- Slightly thins narrow strokes
- Fills tiny gaps after closing

It provides a good balance between noise removal and shape preservation.

e. Risk of a large structuring element

Large structuring elements may

- Remove thin character strokes
- Merge nearby characters
- Distort character shapes
- Reduce recognition accuracy
- Lose fine details

f. Complete preprocessing pipeline

Step 1: Image acquisition (scanner/camera)



Step 2: Grayscale conversion



Step 3: Noise removal (Median filter)



Step 4: Thresholding (Binary conversion)



Step 5: Morphological Opening (remove white noise)



Step 6: Morphological Closing (fill gaps)



Step 7: Character segmentation



Step 8: Normalization (resize and align)



Step 9: Feature extraction (SIFT, SURF, ORB, HOG, or PCA-based features)



Step 10: Classification (e.g., SVM, KNN, CNN, or Neural Network)